A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Marine productivity & fishing - Set 1

Mark scheme

AQA spec 3.5.2.1-3.5.2.2; Biological resources; textbook pp. 338-351

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis **1** In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) **1** The bloom uses up nutrients (falling to 10 by July) – dead plankton sink out of the photic zone carrying nutrients to the seabed **1** So nutrients become limiting in summer despite high light – a smaller autumn bloom follows when storms return nutrients while light is still adequate **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced **1** Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone **1** Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments **1** Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	3
01.4	1 mark for the point below. **1** Phosphates have low solubility and no atmospheric reservoir – they are only added by river runoff or upwelling and are lost to sediments **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Demersal trawl:** seabed species - cod, haddock, plaice, shrimps, scampi – chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass) – high by-catch in mixed fisheries – high fuel use **2 Purse seine:** shoaling pelagic species - tuna, sardines, herring, anchovy – can remove whole shoals – dolphins associated with tuna shoals may be caught **2 Pelagic long-line:** tuna, squid – albatrosses (long-lived, low breeding rate) drown on baited hooks – sharks and turtles caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch **1** Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Lost or discarded nets and traps continue to catch and kill organisms – dead animals act as bait attracting more **1** Biodegradable ropes / panels so lost traps fall apart – radio transmitters on traps so they can be recovered – net recovery schemes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biodegradable** → broken down by decomposers **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 4 points below. **1** k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses **1** Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population **1** Deep-water aggregations easily located and caught by trawling **1** Recovery takes decades so fisheries have had to be closed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young **1** Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort **1** Masks the decline of the breeding population until a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 2 points below. **1** Neither is a target species but both are caught in demersal trawls – skate now extinct in the Baltic and critically endangered elsewhere **1** Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 600 + (90 + 150) - (220 + 80) = 540 **1** Stock fell by 60 so the catch exceeds the MSY / unsustainable **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2
04.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** A + G - M = 90 + 150 - 80 = 160 **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (thousand tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
04.3	**1 mark each** 1 mark for each of the 3 points below. **1** Catch is landed and weighed at ports so it is recorded **1** Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed **1** Representative sampling of a large enough proportion of the area is impractical / expensive – survival of eggs and larvae varies greatly between years **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.4	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2 CPUE:** if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained – a better measure than catch size which technology can prop up **2** Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2.	
05.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2** Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape – square-mesh panels do not close so juveniles escape to grow and breed – TEDs let turtles out – larger mesh sizes generally **2** Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm) – maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) **2** NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats – surplus young spill over and increase surrounding catches – damaged communities recover **2** Closed seasons, ideally in the breeding season, let fish grow and breed – fishermen may oppose because shoaling breeders are easy to catch **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	8
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Pelagic species form single-species shoals so fishing stops when the quota is reached **1** In mixed fisheries many species are caught together – once one species' quota is reached it must be discarded (dead) while fishing for others continues **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
06.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses **1** Setting lines at night when birds are not feeding **1** Bird-scaring streamers / decoys **1** Curved hooks that catch tuna but not birds **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
06.2	**1 mark each** 1 mark for each of the 3 points below. **1 Effort:** limits on boat size, engine power, days at sea, number of licences – decommissioning payments to reduce fleet size where overfishing is unavoidable **1 Methods:** bans or length limits on drift nets – bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) **1** Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **decommissioning** → safely dismantling a power station at the end of its life **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) **1** Juveniles are released when survival chances are good, increasing recruitment to the adult population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.4	**1 mark each** 1 mark for each of the 2 points below. **1** Egg / larval survival varies greatly between years so adult numbers do not predict recruitment **1** Several poor years would lower the MSY – egg surveys allow catches to be reduced in advance rather than after a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **MSY** → the largest catch that can be taken without reducing the stock	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** By-catch = 5.9 + 1.9 = 7.8 t – total = 8.4 t **1** 7.8 / 8.4 x 100 = 92.9% (accept 93%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.2	1 mark for the point below. **1** The nets have a very small mesh size so few non-target animals can escape **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mesh size** → net hole size, set so that young fish escape	1
07.3	**1 mark each** 1 mark for each of the 3 points below. **1** They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young **1** They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred **1** Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young – they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3

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Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Impacts:** population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • **Catch quotas:** limit weight landed – work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • **Gear design:** mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag) – turtle exclusion devices – acoustic pingers for dolphins – curved hooks, decoys, sinkers and night setting to protect albatrosses – biodegradable / radio-tracked traps to reduce ghost fishing • **Restricted effort:** limits on boat size, engine power and days at sea – decommissioning – restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • **Evaluation:** enforcement at sea is difficult – some methods reduce catch or raise costs – quotas need reliable stock data – effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate – as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **quota** → a legal limit on the mass or number that may be caught	9
	• **mesh size** → net hole size, set so that young fish escape • **biodegradable** → broken down by decomposers **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Fishing's impacts go beyond depleting target species: k-selected species that breed late and produce few young collapse and recover slowly; by-catch of immature target fish, over-quota fish and non-commercial species is thrown back dead, with drift nets killing whales, dolphins and turtles, longlines drowning albatrosses and pair trawls porpoises; lost gear keeps ghost fishing; demersal trawls destroy sea fans, corals and seagrass; and removing any species alters its competitors, prey and predators.	
	Catch quotas limit the weight landed and work well for single-species pelagic shoals where fishing stops when the quota is reached, but in mixed demersal fisheries over-quota fish are discarded dead, so the quota does not protect them. Gear design tackles by-catch directly: mesh sizes and square-mesh escape panels let small fish out because, unlike diagonal mesh, they do not close under drag; turtle exclusion devices, acoustic pingers for dolphins, curved hooks, decoys, sinkers and night setting protect turtles, cetaceans and albatrosses; biodegradable and radio-tracked traps limit ghost fishing. Restricting effort - boat size, engine power, days at sea, decommissioning - reduces total pressure, and banning drift nets or demersal trawling on sensitive seabeds such as Rockall Bank protects habitats.	
	Effectiveness depends on enforcement at sea, which is difficult, on reliable stock data for quotas, on the species and fishery type, and on international cooperation because fish migrate; some measures cut catches or raise costs, and as local stocks fall fleets simply move further away. Gear modifications are the most effective per unit cost because they reduce damage without reducing legal catch; quotas and effort limits only work where they are enforced.	
	Weaker answer - Level 2 (5 marks)	
	Fishing damages the environment by overfishing, catching by-catch such as dolphins and damaging the seabed with trawling. Quotas limit how many fish can be caught so stocks can recover. Bigger mesh sizes let small fish escape so they can grow and breed. Turtle exclusion devices let turtles escape from nets. Some fishing methods can be banned in sensitive areas. No-take zones let fish breed. These methods help but they are hard to enforce at sea and some fishermen ignore them.	
	What makes the difference	
	• Level 2 covers impacts and methods soundly but without the mechanisms (why square mesh works, why quotas fail in mixed fisheries).	
	• Level 3 evaluates each method on its own terms and then compares them; Level 2's evaluation is a single general sentence.	
	• Named examples (Rockall Bank, albatross measures, ghost-fishing controls) give Level 3 the range the top band expects.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

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